**PHILOSOPHY 130: PHILOSOPHY OF SPACE AND TIME** Distributed: April 2
**ASSIGNMENT 1: ZENO'S PARADOXES** Due: April 7, in class

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*Reading:* Huggett, *Space from Zeno to Einstein*, Chapter 3 (online as C 1).
*Please write and draw your answers in the space provided.*

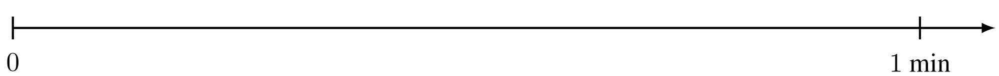
**1.** Fragment 12 in Huggett discusses the **Stadium** paradox, which we did not cover in lecture. It involves three rows of objects (row A stationary, row B moving right, row C moving left), and is meant to show that the assumption of a smallest unit of time leads to a contradiction.

(a) Draw a diagram showing the positions of the three rows at the start and after one "unit" of time. Label each row clearly.

(b) In 1–2 sentences below your diagram, explain what paradoxical conclusion Zeno draws about the relationship between the smallest unit of time and relative motion.

**2. Thomson's Lamp.** Consider a lamp that is toggled on and off at successively halving intervals: on at $t = 0$, off at $t = \frac{1}{2}$ minute, on at $t = \frac{3}{4}$, off at $t = \frac{7}{8}$, and so on. At $t = 1$ minute all the toggles have been completed.

(a) In the timeline below, mark and label the first five toggling times. Write "on" or "off" above each interval.




(b) Is the lamp on or off at $t = 1$? Explain in 1–2 sentences why this is puzzling, and how it relates to the issues raised by the Dichotomy.

**3.** Draw a diagram illustrating a one-to-one correspondence between the points a line segment $[0, 1]$ and a longer one $[0, L]$ (showing how every point in one maps to exactly one point in the other). (Note that there are many ways to do this!) Explain what this diagram shows and why it is relevant to Zeno's paradox of Infinite Divisibility. In particular: if two line segments of different lengths have the same "number" of points, what does this tell us about how length is determined?